ISE 5406: Optimization II

Lecture 6

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Reading for this lecture

- Section 3.4 of Bazaraa et al. (2006)
- Section 3.1 of Boyd and Vandenberghe

Outline

Theorem

Let S be a nonempty convex set in $\mathbb{R}^n$ and $f : S \to \mathbb{R}$ be a convex function. Then for $\bar{\mathbf{x}} \in \text{int}(S)$, there exists a vector $\boldsymbol{\xi}$ such that the hyperplane

$$H = \{(\mathbf{x}, y) : y = f(\bar{\mathbf{x}}) + \boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}})\}$$

supports epi(f) at $[\bar{\mathbf{x}}, f(\bar{\mathbf{x}})]$. In particular, ξ is a subgradient of f at $\bar{\mathbf{x}}$.

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- If f is differentiable at $\bar{\mathbf{x}}$, then $\nabla f(\bar{\mathbf{x}})$ is the *only* subgradient of f at $\bar{\mathbf{x}}$.
- Set of all subgradients of f at $\bar{\mathbf{x}}$ is called the subdifferential $\partial f(\bar{\mathbf{x}}) \subseteq \mathbb{R}^n$.
- $\partial f(\bar{\mathbf{x}})$ is a compact convex set at interior points $\bar{\mathbf{x}}$.

Minimizing a Convex Function

Definition: Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and S be a nonempty convex set in $\mathbb{R}^n$. Then a convex optimization problem is defined by

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \in S. \end{aligned} \tag{1}$$

Theorem

Suppose that $\bar{\mathbf{x}} \in S$ is a local optimal solution to Problem (**??**).

1. Then $\bar{\mathbf{x}}$ is a global optimal solution.
2. If either $\bar{\mathbf{x}}$ is a strict local minimum or f is strictly convex, then $\bar{\mathbf{x}}$ is the unique global optimal solution (it is also a strong local minimum).

Necessary and Sufficient Conditions for a Global Minimum

Theorem

Point $\bar{\mathbf{x}} \in S$ is an optimal solution to Problem (**??**) if and only if f has a subgradient $\boldsymbol{\xi}$ at $\bar{\mathbf{x}}$ such that $\boldsymbol{\xi}^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$.

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Corollary 1: Under the assumptions of the above theorem, if S is an open set (for example, $S = \mathbb{R}^n$), $\bar{\mathbf{x}}$ is an optimal solution to the problem if and only if there exists a zero subgradient of f at $\bar{\mathbf{x}}$.

Corollary 2: Assume that f is differentiable. Then $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$ for all $\mathbf{x} \in S$. Furthermore, if S is open, $\bar{\mathbf{x}}$ is an optimal solution if and only if $\nabla f(\bar{\mathbf{x}}) = \mathbf{0}$.

Examples

- **Solve**

$$\begin{aligned} \min \;\; & x_1 + x_1 x_2 + x_2^2 \\ \text{s.t.} \;\; & (x_1, x_2) \in \mathbb{R}^2. \end{aligned}$$

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Minimizing a Convex Function

Example: $\min\{x^2 : x \in [3,4]\}$. Here $f(x) = x^2$ is a convex function and $S = [3,4]$ is a closed set. However, there does not exist any $\bar{x} \in S$ such that subgradient $\xi = 0$.

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Weierstrass' Theorem (Sufficient Conditions for Existence of Min./Max.)

Let S be a nonempty, compact (closed and bounded) set, and let $f : S \to \mathbb{R}$ be continuous on S. Then the problem $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$ attains its minimum.

Example: Ordinary Least Squares Regression

Solve $\quad \min_{\mathbf{x}\in\mathbb{R}^n} \|A\mathbf{x}-\mathbf{b}\|_2^2 \quad$ given $A \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$.